

Nmap Vulnerability Assessment Lab

Objective: Use Nmap to systematically identify open communication ports, active services, software architecture versions, and potential security vulnerabilities on a designated target environment.

1. Basic Port Scan

COMMAND SYNTAX

```
nmap <target-ip>
```

EXECUTION EXAMPLE

```
nmap 20.244.100.10
```

CORE PURPOSE

- Discover active network hosts within the scope.
- Identify open, closed, and filtered status indicators across standard ports.

2. Service Version Detection

COMMAND SYNTAX

```
nmap -sV <target-ip>
```

EXECUTION EXAMPLE

```
nmap -sV 20.244.100.10
```

CORE PURPOSE

- Query responsive interfaces to determine active daemon software names.
- Extract exact product application version details to check against exploit databases.

3. Operating System Detection

COMMAND SYNTAX

```
sudo nmap -O <target-ip>
```

EXECUTION EXAMPLE

```
sudo nmap -O 20.244.100.10
```

CORE PURPOSE

- Analyze low-level network packet anomalies to identify host operating systems.
- Determine system kernel variants and patch distribution levels.

4. Aggressive Scan

COMMAND SYNTAX

```
sudo nmap -A <target-ip>
```

EXECUTION EXAMPLE

```
sudo nmap -A 20.244.100.10
```

CORE PURPOSE

- Consolidate OS detection, software versioning, and script scanning into a single sequence.
- Gather comprehensive host profiles and network path traceroutes simultaneously.

5. Scan Specific Ports

COMMAND SYNTAX

```
nmap -p 22,80,443 <target-ip>
```

EXECUTION EXAMPLE

```
nmap -p 22,80,443 20.244.100.10
```

CORE PURPOSE

- Isolate exploration exclusively to specified network points.
- Verify functionality and availability on designated service infrastructure.

6. Scan Port Range

COMMAND SYNTAX

```
nmap -p 1-1000 <target-ip>
```

EXECUTION EXAMPLE

```
nmap -p 1-1000 20.244.100.10
```

CORE PURPOSE

- Audit a contiguous custom range index block of communication channels.
- Identify commonly exposed services registered inside conventional standard low boundaries.

7. Vulnerability Scan Using NSE Scripts

COMMAND SYNTAX

```
nmap --script vuln <target-ip>
```

EXECUTION EXAMPLE

```
nmap --script vuln 20.244.100.10
```

CORE PURPOSE

- Leverage Nmap Scripting Engine (NSE) modules to detect known software vulnerabilities.
- Expose insecure platform defaults, weak configuration properties, and missing security updates.

8. Default Script Scan

COMMAND SYNTAX

```
nmap -sC <target-ip>
```

EXECUTION EXAMPLE

```
nmap -sC 20.244.100.10
```

CORE PURPOSE

- Run a standard safe catalog index of default reconnaissance automation scripts.
- Harvest extended metadata strings, protocol banners, and extended application information.

9. Full TCP Port Scan

COMMAND SYNTAX

```
nmap -p- <target-ip>
```

EXECUTION EXAMPLE

```
nmap -p- 20.244.100.10
```

CORE PURPOSE

- Enumerate the entire baseline matrix pool of all 65,535 TCP ports.
- Uncover obfuscated backdoors, hidden services, or private interfaces outside the top-1,000 standard ports.

10. Save Scan Results

COMMAND SYNTAX

```
nmap -sV <target-ip> -oN scan_results.txt
```

EXECUTION EXAMPLE

```
nmap -sV 20.244.100.10 -oN scan_results.txt
```

CORE PURPOSE

- Write raw console output streams to persistent, human-readable text documents.
- Maintain logs for technical analysis evidence, audit records, and reporting documentation.

Regulatory & Operational Compliance Alert

Performing automated network scans, security auditing, and port vulnerability assessments requires explicit, formal authorized rules of engagement from the infrastructure asset owner. Unauthorized scanning can degrade service availability and may violate security compliance policies.